

If list is empty

$\text{if}(\text{head} == \text{NULL})$
 $\text{printf}(\text{"Empty"});$

If pos == 1

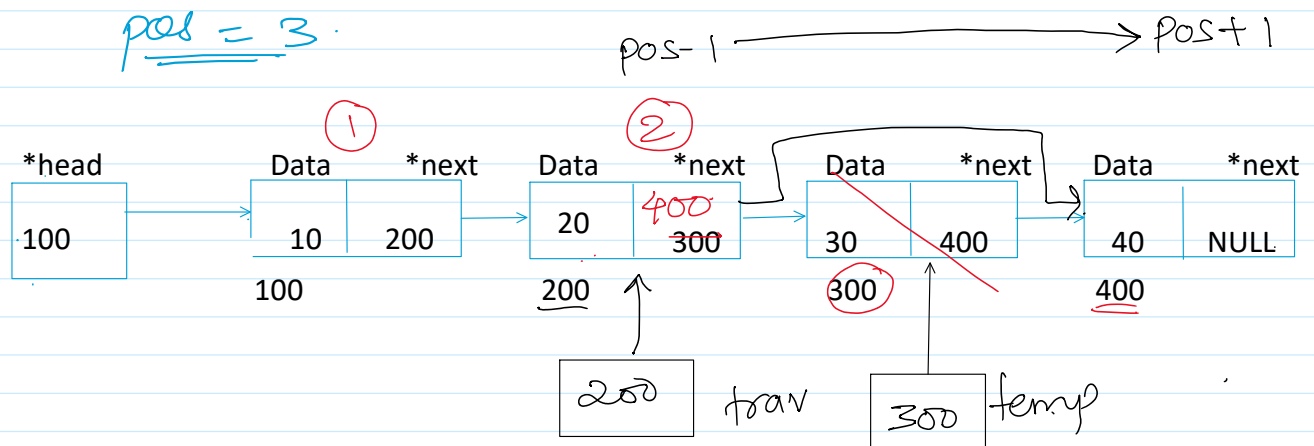
$\text{delete_first_node}();$

$\text{if pos} == \text{count}()$

$\text{delete_last_node}();$

$\text{if pos} < 1$ or $\text{pos} > \text{count}()$
 -3

$\text{printf}(\text{"Cannot delete the node"});$



- 1) Traverse till pos-1 node.
 Struct node *trav = head;

For(int i = 1; i < pos-1; i++)
 Trav = trav->next;

- 2) Take a backup of the pos node.
 Struct node *temp = trav->next;

- 3) Create a link between pos-1 node and pos+1 node.

Pos-1 node --> trav
 Pos node --> temp
 Pos+1 node --> temp->next

$\text{trav} \rightarrow \text{next} = \text{temp} \rightarrow \text{next};$
 $\text{trav} \rightarrow \text{next} = \text{temp} \rightarrow \text{next};$

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Trav->next = temp->next;
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4) Free the temp node.

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Free(temp);
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Temp = NULL;
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Best case : pos = 1 : $O(1)$

Avg/worst case : $O(n)$